chromosomes. When there are three, four, five or six basic sets of chromosomes, they are called triploidy (3x) tetraploidy (4x), pentaploidy (5x) and hexaploidy (6x) respectively. Generally, polyploidy is very common in plants but rarer in animals. An increase in the number of chromosome sets has been an important factor in the origin of new plant species. But higher ploidy level leads to death. Polyploidy is of two types. They are autopolyploidy and allopolyploidy

1. Autopolyploidy

The organism which possesses more than two haploid sets of chromosomes derived from within the same species is called autopolyploid. They are divided into two types. Autotriploids and autotetraploids.

Autotriploids have three set of its own genomes. They can be produced artificially by crossing between autotetraploid and diploid species. They are highly sterile due to defective gamete formation. Example: The cultivated banana are usually triploids and are seedless having larger fruits than diploids. Triploid sugar beets have higher sugar content than diploids and are resistant to moulds. Common doob grass (***Cyanodon dactylon***) is a natural autotriploid. Seedless watermelon, apple, sugar beet, tomato, banana are man made autotriploids.

Autotetraploids have four copies of its own genome. They may be induced by doubling the chromosomes of a diploid species. Example: rye, grapes, alfalfa, groundnut, potato and coffee.

2. Allopolyploidy

An organism which possesses two or more basic sets of chromosomes derived from two different species is called allopolyploidy. It can be developed by interspecific crosses and fertility is restored by chromosome doubling with colchicine treatment. Allopolyploids are formed between closely related species only. (Figure 3.22)

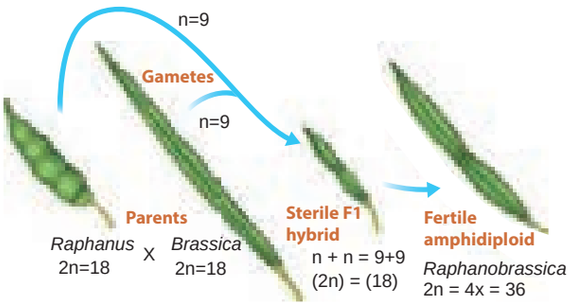


Figure 3. 22 *Rhaphanobrassica*

Example:1 ***Raphanobrassica***, **G.D. Karpechenko** (1927) a Russian geneticist, crossed the radish (***Raphanus sativus***, 2n=18) and cabbage (***Brassica oleracea***, 2n=18) to produce F_1 hybrid which was sterile. When he doubled the chromosome of F_1 hybrid he got it fertile. He expected this plant to exhibit the root of radish and the leaves like cabbage, which would make the entire plant edible, but the case was vice versa, so he was greatly disappointed.

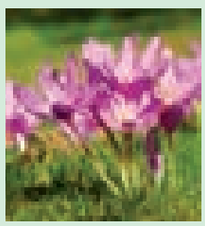
Example: 2 **Triticale**, the successful first man made cereal. Depending on the ploidy level Triticale can be divided into three main groups.

(i). Tetraploidy: Crosses between diploid wheat and rye.

(ii). Hexaploidy: Crosses between tetraploid wheat Triticum durum (macaroni wheat) and rye

(iii). Octoploidy: Crosses between hexaploid wheat T. aestivum (bread wheat) and rye

Hexaploidy Triticale hybrid plants demonstrate characteristics of both macaroni wheat and rye. For example, they combine the high-protein content of wheat with rye's high content of the amino acid lysine, which is low in wheat. It can be explained by chart below (Figure: 3.23).



Colchicine , an alkaloid is extracted from root and corms of ***Colchicum autumnale***, when applied in low concentration to the growing tips of the plants it will induce polyploidy. Surprisingly it does not affect the source plant *Colchicum*, due to presence of anticolchicine.

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